

Admissible Reconstruction

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Abstract

This monograph isolates and proves four theorems governing when structure survives transformation, and when it does not. Each theorem is derived directly from the admissibility apparatus developed elsewhere in this research program — the relation $x \rightsquigarrow_R y$, the reachable set $R(x)$, the time-indexed admissibility structure R_t , and the closure functional $C(X) = -\log \mu(R(X))$ — rather than introduced as free-standing results. *Representation Invariance* states that two representations are semantically equivalent whenever they induce the same admissibility structure, independent of their physical substrate. *Admissibility Preservation* shows that a trajectory’s coherence is not an independent postulate but a consequence of the admissibility relation failing to degenerate: coherence just is the nontriviality of $R(S_t)$ maintained across a trajectory. *Monotonic Relaxation* identifies the precise, conditional sense in which a scalar measure of inadmissibility decreases under repair dynamics specifically, and distinguishes this claim carefully from the unrelated ordering relation of monotonic continuation. *Non-Identifiability* shows that the existence of histories $H_1 \neq H_2$ with $O(H_1) = O(H_2)$ is not a pathological edge case but a structural consequence of any observation map that preserves only reachable-set structure rather than full state — the same fact that elsewhere in this program is called the restorability boundary, here proven as a limitative theorem in its own right.

Together the four results describe a single underlying architecture: representation, coherence, repair, and the limits of recovery are not four separate phenomena requiring four separate theories, but four faces of one admissibility relation examined at different points in its behavior.

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Notation Index

$x \rightsquigarrow_R y$	x admissibly continues to y under admissibility structure R .
$R(x)$	the reachable set of x : $\{ y : x \rightsquigarrow_R y \}$.
R_t	the admissibility structure indexed by time t ; the static relation R is the frozen special case $R_t \equiv R$.
$S_t \leq S_{t+1}$	monotonic continuation: S_{t+1} preserves the distinctions of S_t that remain necessary for the reachable trajectory. Ordering, not magnitude.
$C(X) = -\log \mu(R(X))$	the closure functional; a scalar measure of inadmissibility, larger when fewer futures remain reachable from X .
$G(X) = \log \mu(R(X))$	the generativity measure; $G(X) = -C(X)$.
H	a complete history (ground truth), as retained by a simulator or, more generally, as the full state trajectory prior to any lossy observation.
O_t	the observable projection of H available at time t ; a deliberately lossy map $O : H \rightarrow O_t$.
$\hat{H}$	a reconstruction of H inferred from O_t alone, without privileged access to H .
φ	an admissibility-isomorphism between two representations: a map such that $x \rightsquigarrow_R y \iff \varphi(x) \rightsquigarrow_{R'} \varphi(y)$.

Chapter 1

Introduction

This monograph is organized around a single question: when does structure survive transformation, and when does it not? The question has appeared throughout this research program under several names — reconstruction, repair, restorability, continuation — and has been approached from several directions, including a formal treatment of admissibility and irreversibility, a foundational account of observability as restorability, and an extended study of continuation across unequal timescales. What follows is not a new direction but a consolidation: four theorems, each of which has appeared informally or partially elsewhere in this program, stated here as precise results and derived directly from the admissibility apparatus rather than argued for case by case.

The apparatus itself is simple to state. An admissibility structure R determines, for any state x , a reachable set $R(x)$ of states y such that $x \rightsquigarrow_R y$. Nothing about R is assumed beyond this: it is a relation, not a metric, and it need not be symmetric, transitive, or total. Historical work in this program has shown that a great deal of structure can be recovered from this minimal starting point once it is allowed to vary with time, R_t , and once transitions between admissible structures are themselves treated as admissible or inadmissible events rather than as external clock ticks.

The four theorems developed in this monograph correspond to four questions one can ask of this apparatus. First: when are two representations, possibly built from entirely different physical or computational substrates, the same object as far as R is concerned? This is Representation Invariance, the subject of Chapter 2. Second: what does it mean for a trajectory through admissibility structure to be coherent, and is coherence an additional postulate or a consequence of the relation itself? This is Admissibility Preservation, the subject of Chapter 3. Third: under what conditions, and in what precise sense, does a scalar measure of inadmissibility decrease along a trajectory, and how does this relate to — and how must it be distinguished from — the ordering relation of monotonic continuation developed elsewhere in this program? This is Monotonic Relaxation, the subject of Chapter 4. Fourth: when is it structurally impossible, not merely difficult, to recover which of two histories actually occurred from the evidence either would have left behind? This is Non-Identifiability, the subject of Chapter 5.

A methodological note is appropriate before proceeding. Each of the four theorems could be, and in earlier work occasionally has been, treated as an independent claim requiring its own justification. The approach taken here is different. Every theorem below is

derived from definitions already established elsewhere in this program — the admissibility relation, the reachable set, the closure functional — so that what is new in this monograph is not the vocabulary but the demonstration that four apparently distinct phenomena are, in fact, four views of one relation. Where a theorem’s content is close to definitional, this is stated plainly rather than disguised as a deeper result; where a theorem requires a genuine argument, as with Admissibility Preservation and Monotonic Relaxation, the argument is given in full and its conditions of validity are stated explicitly rather than left implicit.

Chapter 2

Representation Invariance

2.1 Statement

The question addressed in this chapter is when two representations, possibly realized in entirely different substrates, ought to be considered the same object. The answer given here is that sameness of substrate is irrelevant; what matters is sameness of admissibility structure.

Definition 2.1 (Admissibility-isomorphism). Let R be an admissibility structure on a set X and R' an admissibility structure on a set X' . A map $\varphi : X \rightarrow X'$ is an *admissibility-isomorphism* if, for all $x, y \in X$,

$$x \rightsquigarrow_R y \iff \varphi(x) \rightsquigarrow_{R'} \varphi(y).$$

Theorem 2.2 (Representation Invariance). *If there exists an admissibility-isomorphism $\varphi : X \rightarrow X'$ between (X, R) and (X', R') , then (X, R) and (X', R') are semantically equivalent, regardless of whatever physical or computational substrate realizes X and X' .*

2.2 Discussion

The theorem as stated is close to definitional, and this is by design rather than by accident. Semantic equivalence, within this research program, has never been defined as identity of physical encoding; it has been defined, throughout the treatment of admissible representation and context preservation developed elsewhere in this program, as identity of what a representation permits and forbids as continuation. Theorem 2.2 simply makes that definition explicit as an invariance statement rather than leaving it distributed across the apparatus that depends on it.

What is not definitional, and what does real work when the theorem is applied, is establishing that a particular φ is in fact an admissibility-isomorphism between two representations of interest. This is generally the harder direction: showing that a biphasic transformation — structured meaning realized as continuous speech, or as sequential text, or as embodied gesture — preserves the reachable-set structure exactly, rather than merely preserving some coarser statistic that happens to look similar under casual inspection. The

content of a representation-invariance claim, in any particular application, is carried entirely by the verification that φ satisfies the biconditional in both directions: that admissible continuations in one representation correspond exactly to admissible continuations in the other, with neither representation admitting a continuation the other forbids.

Remark 2.3. It follows immediately from Theorem 2.2 that representation invariance is an equivalence relation on admissibility structures: the identity map witnesses reflexivity, the inverse of an admissibility-isomorphism is again an admissibility-isomorphism, giving symmetry, and composition of admissibility-isomorphisms is again an admissibility-isomorphism, giving transitivity. Consequently, the class of representations equivalent to a given (X, R) partitions into a single equivalence class under φ , and asking whether two representations are “the same” reduces to asking whether they lie in the same class.

Corollary 2.4. *No property of X or X' that is not preserved by every admissibility-isomorphism can be a semantic property of the representation. In particular, substrate-specific properties — the particular physical channel through which a representation is realized, its encoding cost, or its production latency — are not semantic properties under this framework unless they happen to be expressible as constraints on R itself.*

Corollary 2.4 is worth stating explicitly because it draws a sharp boundary that is easy to blur in practice. Structural semantics, as developed elsewhere in this program, is precisely the study of what survives this partition; anything that does not survive it belongs to a different, and legitimate, subject — the study of channel, cost, and implementation — but not to the study of what a representation *means*.

Chapter 3

Admissibility Preservation

3.1 Statement

The question addressed in this chapter is what it means for a trajectory through admissibility structure to remain coherent. The answer given here is that coherence is not an additional condition imposed on top of admissibility; it is what nontriviality of the admissibility relation amounts to when examined along a trajectory rather than at a single state.

Definition 3.1 (Trajectory). A trajectory is a sequence of states $S_0, S_1, \dots, S_t$ such that $S_i \rightsquigarrow_R S_{i+1}$ for each $i < t$, where R may itself vary with i if the underlying admissibility structure is time-indexed as R_i .

Definition 3.2 (Degeneracy). A trajectory degenerates at step i if $R(S_i)$ is empty (no continuation is admissible) or a singleton entirely determined by $S_0, \dots, S_{i-1}$ (exactly one continuation is admissible, and it was already fixed before S_i was reached).

Theorem 3.3 (Admissibility Preservation). *A trajectory $S_0, \dots, S_t$ is coherent if and only if it does not degenerate at any step $i \leq t$; equivalently, if and only if $|R(S_i)|$ exceeds a nontrivial floor relative to what has already been committed by $S_0, \dots, S_{i-1}$, at every step.*

Proof. Coherence, as the term is used throughout this program, has no content apart from the admissibility relation itself: to say a trajectory is coherent is to say that at each step, the relation $x \rightsquigarrow_R y$ continues to discriminate admissible from inadmissible continuations non-vacuously. Suppose the trajectory degenerates at step i because $R(S_i) = \emptyset$. Then no y satisfies $S_i \rightsquigarrow_R y$, and the relation, restricted to continuations from S_i , is uniformly false; there is no sense in which the trajectory can be said to continue admissibly past S_i , coherently or otherwise, since no continuation is admissible at all. Suppose instead the trajectory degenerates because $R(S_i)$ is a singleton already fixed by $S_0, \dots, S_{i-1}$. Then the relation restricted to continuations from S_i is uniformly true of exactly one candidate and uniformly false of every other; the admissibility relation is doing no discriminating work at step i , since its outcome was determined before S_i was reached, and the statement " $S_i \rightsquigarrow_R S_{i+1}$ is coherent" becomes true independently of anything about S_i itself. In either degenerate case, the relation $\rightsquigarrow_R$ fails to be a nontrivial constraint at step i : it is either vacuously unsatisfiable or vacuously predetermined. Conversely, if $R(S_i)$ exceeds a nontrivial floor at every step

— if there is always more than one admissible continuation not already fixed by the trajectory’s own history — then the admissibility relation is doing genuine discriminating work at every step, which is what coherence consists in. This establishes the biconditional. $\square$

3.2 Discussion

The proof above is deliberately minimal, and its minimality is the point. Coherence has sometimes been treated, informally, as though it were an additional property a trajectory could have or lack independent of the admissibility relation that generates it — as though a trajectory could be admissible at every step and yet somehow incoherent, or coherent despite violating admissibility. Theorem 3.3 rules this out: coherence and non-degeneracy of the admissibility relation are the same property under two names. This matters directly for the definition of monotonic continuation developed elsewhere in this program, where the ordering $S_t \leq S_{t+1}$ presupposes, without stating, that $R(S_t)$ remains nontrivial; Theorem 3.3 supplies the missing lemma that makes this presupposition explicit and shows exactly what fails when it does not hold.

Corollary 3.4. *Any operational test for coherence — such as a minimum support ratio required of a discourse trajectory before it is judged coherent — is correctly understood as an operational proxy for $|R(S_i)|$ remaining above a nontrivial floor, not as an independent criterion. Where such a proxy diverges from the underlying reachable-set condition, the proxy, not the theorem, should be revised.*

Corollary 3.4 is included because domain-specific coherence tests — support ratios, consistency scores, and similar quantities — are frequently introduced as though they were primitive definitions of coherence in their own right. Theorem 3.3 reduces the space of legitimate such tests to those that track, however indirectly, the nondegeneracy of the admissibility relation, and gives a principled way to evaluate a proposed test: ask whether it can be false while $R(S_i)$ remains nontrivial, or true while $R(S_i)$ has degenerated. Either case indicates the proxy has diverged from what coherence actually consists in.

Chapter 4

Monotonic Relaxation

4.1 A Terminological Warning

This chapter's theorem shares a name, and very little else, with monotonic continuation as developed elsewhere in this program. The two should not be conflated, and the distinction is stated here before the theorem itself to prevent exactly that conflation. Monotonic continuation is an ordering relation, $S_t \leq S_{t+1}$, holding between successive states of a trajectory and asserting that later states preserve the distinctions from earlier states that remain necessary for the reachable continuation; it says nothing about any scalar quantity increasing or decreasing. Monotonic relaxation, the subject of this chapter, is a claim that a specific scalar functional decreases along a specific class of trajectories. The two claims are compatible, and one will be shown below to imply a restricted version of the other under an additional hypothesis, but they are not restatements of each other and a trajectory can satisfy either without satisfying the other.

4.2 Statement

Definition 4.1 (Closure functional). For a state X , define $C(X) = -\log \mu(R(X))$, where μ is a measure on the state space assigning nonzero mass to any nonempty reachable set. $C(X)$ is large when few futures remain reachable from X and small when many do; it is a scalar measure of inadmissibility.

Definition 4.2 (Repair dynamics). A trajectory follows *repair dynamics* if each transition $S_i \rightarrow S_{i+1}$ is selected, among admissible continuations, to increase reachable-set volume: $\mu(R(S_{i+1})) \geq \mu(R(S_i))$. This is the operational content of repair within this program's taxonomy of admissibility-modifying operations: repair is the restorative case, distinguished from constitutive and regulatory modification precisely by this volume-increasing character.

Theorem 4.3 (Monotonic Relaxation). *If a trajectory $S_0, \dots, S_t$ follows repair dynamics throughout, then $C(S_i)$ is non-increasing along the trajectory: $C(S_{i+1}) \leq C(S_i)$ for every $i < t$.*

Proof. By definition of repair dynamics, $\mu(R(S_{i+1})) \geq \mu(R(S_i))$ for every $i < t$. Since

$-\log$ is a strictly decreasing function on positive reals, $\mu(R(S_{i+1})) \geq \mu(R(S_i))$ implies $-\log \mu(R(S_{i+1})) \leq -\log \mu(R(S_i))$, that is, $C(S_{i+1}) \leq C(S_i)$. $\square$

4.3 Discussion

The proof is immediate once repair dynamics are defined as volume-increasing, and this is deliberate: the theorem's content is not in the inference from the definition, which is a one-line consequence of $-\log$ being decreasing, but in the definition itself being the correct operational characterization of repair. The definition is not stipulated arbitrarily; it is inherited directly from the restorative case of this program's modificatory-operation taxonomy, where repair is already characterized, independently of this chapter, as the operation type that restores lost admissible continuations. Theorem 4.3 shows that once repair is characterized this way, monotone decrease of $C(\cdot)$ follows automatically; it is not an additional empirical claim about repair processes but a restatement of what "restoring lost continuations" already meant, now expressed as a statement about a scalar functional.

Remark 4.4 (Scope of the theorem). The hypothesis is essential and is not automatically satisfied by an arbitrary admissible trajectory. Constitutive and regulatory modifications — the other two branches of this program's taxonomy of operations on admissibility structure — need not increase $\mu(R(\cdot))$, and ordinary state evolution under a fixed R_t can move toward states with strictly smaller reachable sets without this constituting any failure of admissibility. A claim of the form "inadmissibility always decreases" would be false; Theorem 4.3 is true only of the restorative subclass of trajectories, and its scope should be stated with this restriction every time it is invoked.

Corollary 4.5 (Conditional relation to monotonic continuation). *If a trajectory follows repair dynamics and additionally satisfies $R(S_{i+1}) \supseteq^* R(S_i)$ in the starred sense used by monotonic continuation — that is, S_{i+1} retains the admissible continuations of S_i that remain significant for the reachable trajectory, rather than merely having a reachable set of at least equal measure — then the trajectory satisfies both $S_i \leq S_{i+1}$ in the sense of monotonic continuation and $C(S_{i+1}) \leq C(S_i)$ in the sense of this chapter. Neither condition implies the other in general: a trajectory can increase $\mu(R(\cdot))$ while failing to retain the specific significant continuations monotonic continuation requires, by replacing them with an equal or greater volume of differently located reachable states; and a trajectory can satisfy monotonic continuation's retention condition while $\mu(R(\cdot))$ decreases, if what is retained is a shrinking but still significant core.*

Corollary 4.5 is the precise statement of the relationship gestured at in the opening warning of this chapter. The two notions of monotonicity coincide only under an additional hypothesis that must be checked independently in any given application, and the corollary exists chiefly to prevent the two theorems from being cited interchangeably.

Chapter 5

Non-Identifiability

5.1 Statement

The question addressed in this chapter is when it is impossible, not merely difficult, to recover which of two histories actually occurred from the evidence either would have left behind. The elementary direction of this claim is close to definitional; the substantive direction is showing that the relevant class of observation maps is genuinely, structurally non-injective, rather than merely non-injective in unusual or contrived cases.

Definition 5.1 (Observation map). An observation map is a function $O : H \rightarrow O_t$ from complete histories to observables, where O_t is whatever evidence remains available at a given time to an observer without access to H itself.

Theorem 5.2 (Non-Identifiability, elementary direction). *If $H_1 \neq H_2$ and $O(H_1) = O(H_2)$, then no function $f : O_t \rightarrow H$ satisfies $f(O(H_1)) = H_1$ and $f(O(H_2)) = H_2$ simultaneously.*

Proof. Suppose such an f existed. Since $O(H_1) = O(H_2)$, write $o = O(H_1) = O(H_2)$. Then $f(o) = f(O(H_1)) = H_1$ and also $f(o) = f(O(H_2)) = H_2$, so $H_1 = H_2$, contradicting the hypothesis $H_1 \neq H_2$. $\square$

This direction is elementary because it is a direct consequence of f being required to be a function, together with the assumed collision $O(H_1) = O(H_2)$; no property of admissibility structure is needed to prove it. What requires the admissibility apparatus developed elsewhere in this program is establishing that such collisions are not pathological exceptions confined to contrived cases, but a structural feature of the observation maps that actually arise when observation is constrained to preserve only reachable-set structure.

Theorem 5.3 (Non-Identifiability, structural direction). *Let O be any observation map that factors through the reachable-set structure — that is, suppose $O(H) = \psi(R(H))$ for some function ψ , so that O records only which continuations remain admissible from H , not H itself. Then whenever two histories $H_1 \neq H_2$ satisfy $R(H_1) = R(H_2)$, it follows that $O(H_1) = O(H_2)$, and Theorem 5.2 applies. Moreover, this is not a degenerate case: $R(H_1) = R(H_2)$ is precisely the condition under which H_1 and H_2 lie in the same equivalence class of the admissibility relation, and any admissibility structure with a non-trivial quotient — that is, any admissibility structure in which more than one history maps to the same reachable set — guarantees the existence of such collisions.*

Proof. The first claim is immediate: if $R(H_1) = R(H_2)$ then $\psi(R(H_1)) = \psi(R(H_2))$ trivially, since ψ is a function of the reachable set alone. For the second claim, recall that $R(\cdot)$ partitions the space of histories into equivalence classes by definition: two histories lie in the same class exactly when they have the same reachable set. An admissibility structure has a trivial quotient only when this partition is the discrete partition, that is, only when no two distinct histories ever share a reachable set. This is not the generic case for admissibility structures of the kind developed elsewhere in this program, where reachable sets are typically determined by a bounded amount of structure — the admissible continuations consistent with what has been committed so far — rather than by the full, unbounded historical record. Whenever the reachable set is a function of bounded information about H while H itself can vary over an unbounded space, the pigeonhole principle guarantees histories differing in the unrecorded portion while agreeing in the recorded portion, hence sharing a reachable set, hence colliding under O . $\square$

5.2 Discussion

Theorem 5.3 is the precise sense in which Non-Identifiability is not a separate result requiring separate machinery, but the restorability boundary developed elsewhere in this program, proven here as a limitative theorem in its own right rather than argued for informally. The restorability boundary has previously been discussed in terms of what invariants survive repeated reconstruction and what information is irrecoverably lost; Theorem 5.3 identifies exactly the mechanism by which loss becomes structurally guaranteed rather than merely likely: any observation map constrained to record reachable-set structure, applied to a space of histories whose reachable sets are determined by less information than the histories themselves contain, is non-injective as a matter of definition, not as a matter of contingent bad luck in any particular case.

Corollary 5.4. *Given an observed collision $O(H_1) = O(H_2)$ with $H_1 \neq H_2$, the question of why reconstruction fails for this pair admits a principled decomposition. Either $R(H_1) \neq R(H_2)$, in which case the observation map O has discarded information that a less lossy observation map could in principle have retained, and the failure is a failure of the particular observation procedure; or $R(H_1) = R(H_2)$, in which case no observation map factoring through reachable-set structure can distinguish H_1 from H_2 , and the failure is a failure of recoverability as such, not of any particular reconstruction procedure.*

Corollary 5.4 gives the precise form of the question that motivates this entire research program's treatment of reconstruction: whether a given case of reconstruction failure reflects an inference that was performed incorrectly, or evidence that has structurally ceased to exist. The corollary shows this is not merely a useful distinction to draw informally but a genuine dichotomy, decidable in principle by checking whether the colliding histories share a reachable set.

Chapter 6

Conclusion

The four theorems of this monograph were introduced, in Chapter 1, as four questions one can ask of a single admissibility relation. Having proven each in turn, it is worth stating plainly what has and has not been established.

Representation Invariance (Theorem 2.2) established that sameness of representation, properly understood, is sameness of admissibility structure and nothing else; substrate is invisible to the theorem except insofar as it constrains R . Admissibility Preservation (Theorem 3.3) established that coherence of a trajectory is not a separate property requiring its own definition, but exactly the nondegeneracy of the admissibility relation examined along that trajectory. Monotonic Relaxation (Theorem 4.3) established that a scalar measure of inadmissibility decreases along repair trajectories specifically, as a direct consequence of how repair is already characterized elsewhere in this program, and Corollary 4.5 fixed the precise, non-automatic relationship between this claim and the unrelated ordering relation of monotonic continuation. Non-Identifiability (Theorem 5.3) established that failures of historical reconstruction are not contingent misfortunes but a structural guarantee whenever observation is constrained to reachable-set structure and the space of possible histories exceeds what that structure can encode, and Corollary 5.4 gave a principled way to diagnose, in any particular case, whether a reconstruction failure reflects a correctable error or a genuine limit.

Two things follow from having stated these four theorems together rather than separately, as they have variously appeared in earlier work in this program. The first is economy: each theorem's proof drew only on definitions already established elsewhere, and none required introducing new primitive vocabulary. The second, and more important, is that the four theorems are visibly not four independent facts about four different subjects — representation, coherence, repair, and reconstruction — but four consequences of one relation, $x \rightsquigarrow_R y$, examined respectively at a single state, along a trajectory, under a restorative dynamics, and under a lossy projection. What varies across the four chapters is not the object of study but the operation performed on it. This is, in the end, the thesis this monograph exists to make precise: that admissibility is not one tool among several available for studying structure under transformation, but the common substrate from which invariance, coherence, repair, and the limits of recovery are all derivable in turn.

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